



SAC

2025

André Pereira

Top HPC systems in TOP500 lists
(some images are borrowed)



What is TOP500?



TOP500

From Wikipedia, the free encyclopedia

The **TOP500** project ranks and details the 500 most powerful non-distributed computer systems in the world. The project was started in 1993 and publishes an updated list of the supercomputers twice a year. The first of these updates always coincides with the International Supercomputing Conference in June, and the second is presented at the ACM/IEEE Supercomputing Conference in November. The project aims to provide a reliable basis for tracking and detecting trends in high-performance computing and bases rankings on HPL,^[1] a portable implementation of the high-performance LINPACK benchmark written in Fortran for distributed-memory computers.

The 60th TOP500 was published in November 2022. Since June 2022, the United States' Frontier is the most powerful supercomputer on TOP500, reaching 1102 petaFlops (1.102 exaFlops) on the LINPACK benchmarks.^[2] The United States has by far the highest share of total computing power on the list (nearly 50%),^[3] while China currently leads the list in number of systems with 173 supercomputers, with the U.S. not far behind in second place.

The TOP500 list is compiled by Jack Dongarra of the University of Tennessee, Knoxville, Erich Strohmaier and Horst Simon of the National Energy Research Scientific Computing Center (NERSC) and Lawrence Berkeley National Laboratory (LBNL), and, until his death in 2014, Hans Meuer of the University of Mannheim, Germany.

The TOP500 project lists also Green500 and HPCG benchmark list.

TOP500

TOP 500

The List.

Key people	Erich Strohmaier Jack Dongarra Horst Simon Martin Meuer
Established	24 June 1993
Website	www.top500.org

LINPACK benchmarks (HPL)



LINPACK benchmarks

From Wikipedia, the free encyclopedia

For the software library, see [LINPACK](#).

The **LINPACK Benchmarks** are a measure of a system's **floating point** computing power. Introduced by **Jack Dongarra**, they measure how fast a computer solves a dense n by n **system of linear equations** $Ax = b$, which is a common task in **engineering**.

The latest version of these **benchmarks** is used to build the **TOP500** list, ranking the world's most powerful supercomputers.^[1]

The aim is to approximate how fast a computer will perform when solving real problems. It is a simplification, since no single computational task can reflect the overall performance of a computer system. Nevertheless, the LINPACK benchmark performance can provide a good correction over the peak performance provided by the manufacturer. The peak performance is the maximal theoretical performance a computer can achieve, calculated as the machine's frequency, in cycles per second, times the number of operations per cycle it can perform. The actual performance will always be lower than the peak performance.^[2] The **performance of a computer** is a complex

HPL, written in C, measures the sustained floating-point rate (GFLOPs/s) to solve a dense system of linear equations using double-precision floating-point arithmetic

LINPACK benchmarks

Original author(s)	Jack Dongarra , Jim Bunch, Cleve Moler, and Gilbert Stewart
Initial release	1979
Website	www.netlib.org/benchmark/hpl/ 

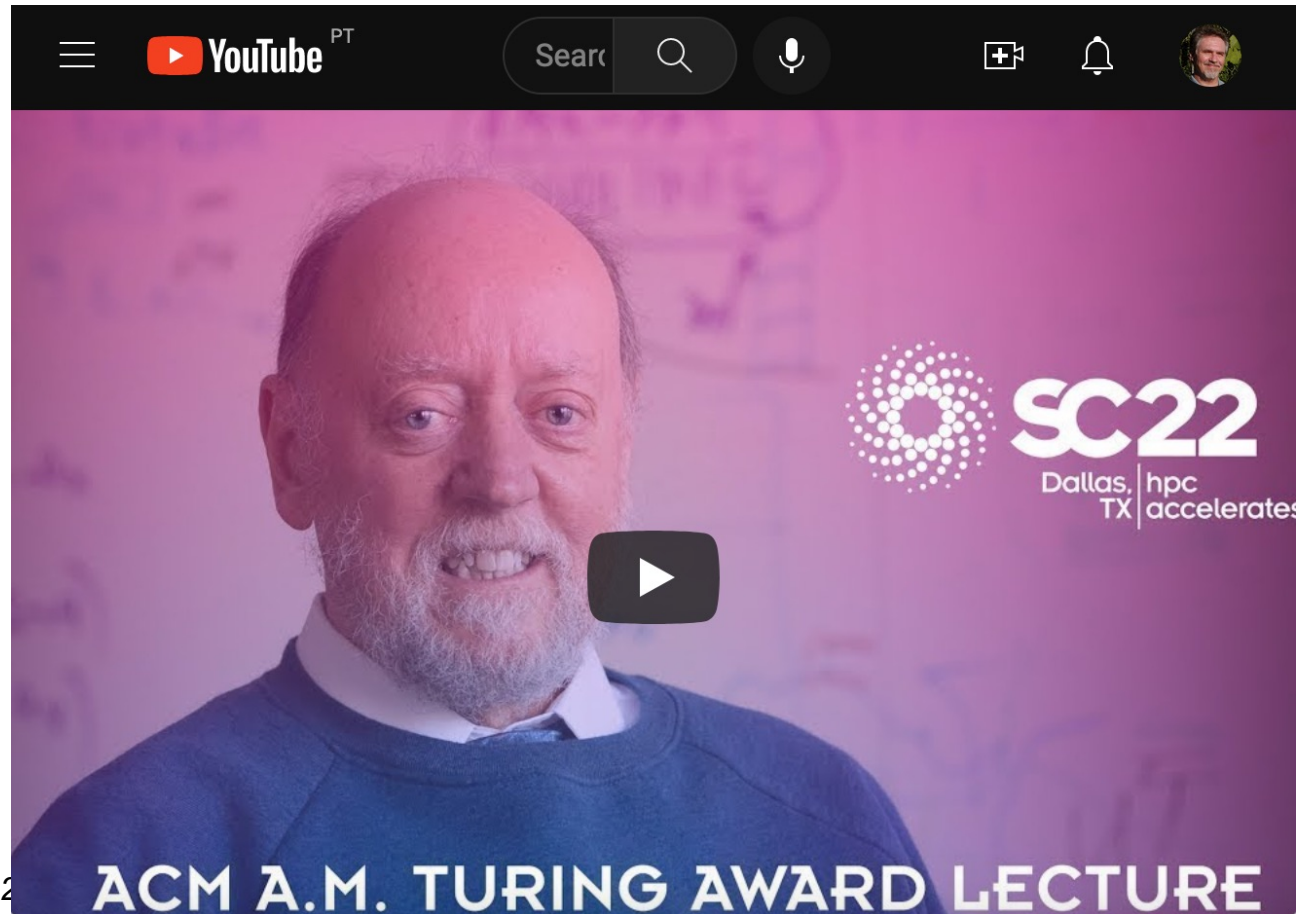
For pioneering contributions
to numerical algorithms and
libraries that enabled high
performance computational
software to keep pace with
exponential hardware
improvements

JACK DONGARRA



AWARD
2021
A.M.
TURING

ACM Turing Award 2021: Jack Dongarra



APereira, AJProença, SAC, 2

<https://www.youtube.com/watch?v=lsnRP9akCDk>



1. TOP500 (*LINPACK*)

- a) TOP10 lists from Nov'19 to Nov'23
- b) Analysis of some relevant systems and architectures
- c) Country distribution over the past 25 years
- d) PU chip technology evolution in the past 25 years & since last year
- e) Evolution of the accelerators since they were available

2. GREEN500

- a) TOP5 list from Nov'19 to Nov'22
- b) Analysis of some relevant systems and architectures

3. HPCG

- a) HPCG vs. HPL: an overview
- b) TOP10 in Nov'22

4. HPL-AI

- a) High-performance Linpack (HPL) and artificial intelligence (AI) workloads

Top 10 HPC systems Nov'23 TOP500

Since June 22

New in the TOP10 list,
with 4th Gen Xeon & Intel GPU

New in the TOP10 list,
with 4th Gen Xeon & H100

3 EuroHPC systems in
the TOP10 list!

Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)		
1	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,699,904	1,194.00	1,679.82	22,703		
2	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	4,742,808	585.34	1,059.33	24,687		
3	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Microsoft Azure United States	1,123,200	561.20	846.84			
4	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899		
5	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107		
6	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy	1,824,768	238.70	304.47	7,404		
7	Summit - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	48.60				
8	MareNostrum 5 ACC - BullSequana XH3000, Xeon Platinum 8460Y+ 40C 2.3GHz, NVIDIA H100 64GB, Infiniband NDR200, EVIDEN EuroHPC/BSC Spain	680,960	138.20				
9	Eos NVIDIA DGX SuperPOD - NVIDIA DGX H100, Xeon Platinum 8480C 56C 3.8GHz, NVIDIA H100, Infiniband NDR400, Nvidia NVIDIA Corporation United States			485,888	121.40	188.65	
10	Sierra - IBM Power System AC922, IBM POWER9 22C 3.1GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM / NVIDIA / Mellanox DOE/NNNSA/LLNL United States			1,572,480	94.64	125.71	7,438

Top 10 HPC systems

Nov'23 TOP500

Since June 22

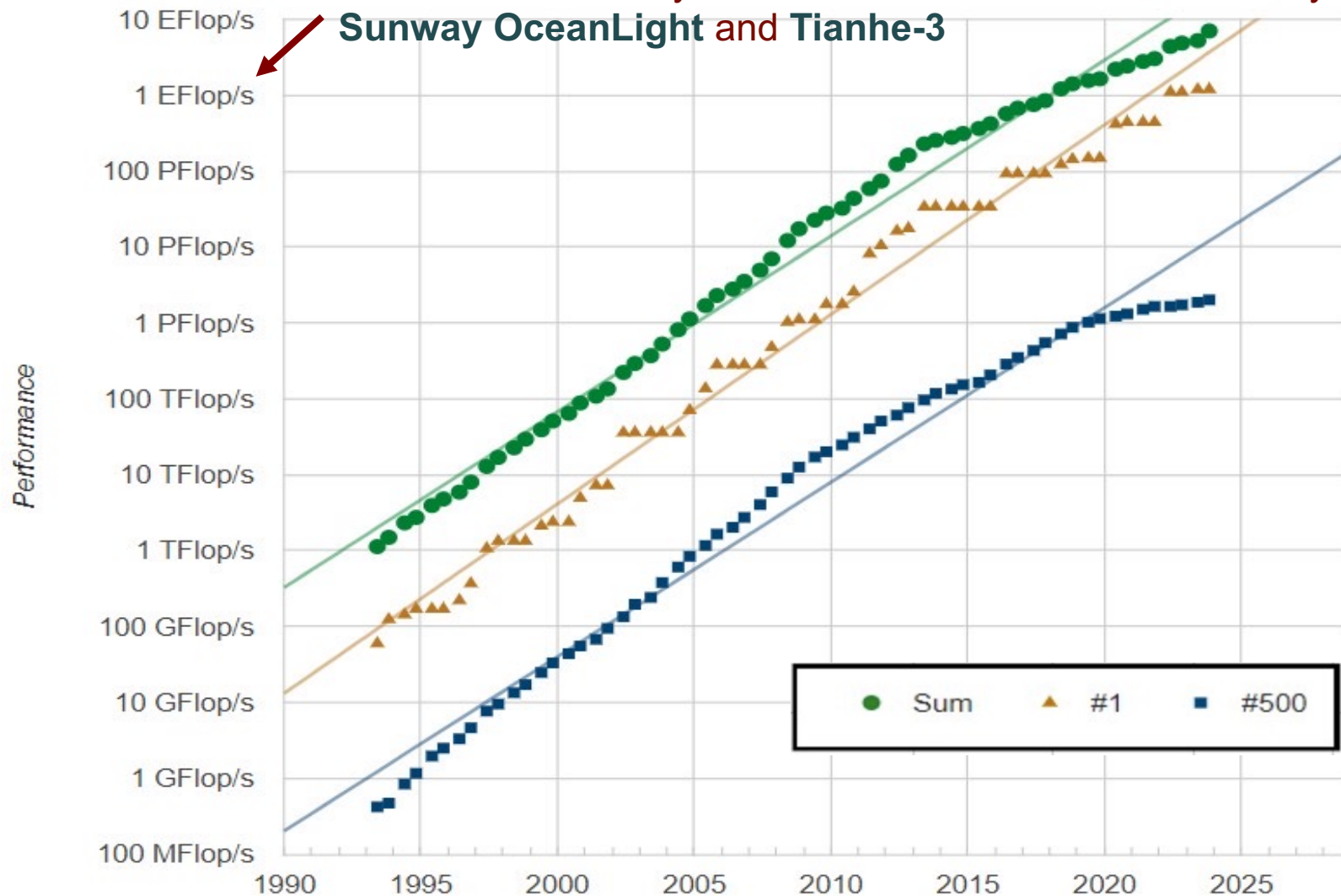
New in the TOP10 list, with 4th Gen Xeon & Intel GPU

New in the TOP10 list, with 4th Gen Xeon & H100

3 EuroHPC systems in the TOP10 list!



The Chinese already overcame this barrier in 2021 w/ 2 systems:
Sunway OceanLight and Tianhe-3



Analysis of some key systems in 2024



1. **#1** since Nov'22: **Frontier** (*AMD Epyc Trento 64c + Instinct MI250x*)
2. **#2** in Nov'22 (#1 in Jun'20): **Fugaku** (*Fujitsu A64FX, 48 cores*)
3. **#4** in Nov'22: **Leonardo**, (*3rd Gen Xeon, 32c + NVidia Ampere A100*)
4. **#5** in Nov'22 (#1 in Nov'18): **Summit** (*IBM POWER9, 22 cores + NVidia Volta GV100*)

1



Frontier at ORNL



Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE
DOE/SC/Oak Ridge National Laboratory
United States
Nov'23

Competitive procurement asking for:

50–100× application performance of Titan

Support for traditional modeling and simulation, high-performance data analysis, and artificial intelligence applications

Peak performance >1000 PF

Smooth transition for existing and future applications



Jaguar: 2.3 PF
World's Fastest

2008



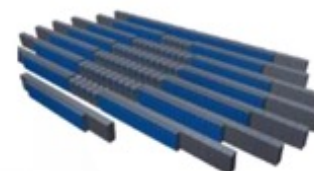
Titan: 27 PF
Accelerated Computing
World's Fastest

2012



Summit: 200 PF
Accelerated Computing
5–10× Titan Performance

2017



FRONTIER

Frontier: >1000 PF
Competitive Procurement
5-10× Summit
Performance

2021



JAGUAR

No. 1 in 2009, 2010



TITAN

No. 1 in 2012



SUMMIT

No. 1 in 2018



FRONTIER

No. 1 in 2022

Frontier Overview Extraordinary Engineering



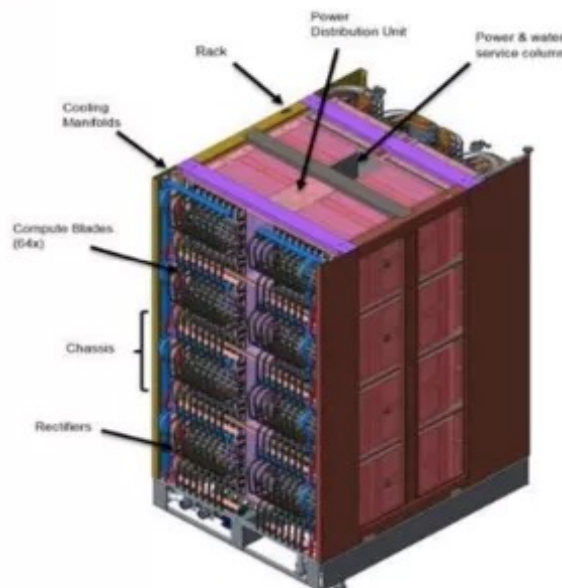
System

- 2 EF Peak DP FLOPS
- 74 compute racks
- 29 MW Power Consumption
- 9,408 nodes
- 9.2 PB memory
(4.6 PB HBM, 4.6 PB DDR4)
- Cray Slingshot network with dragonfly topology
- 37 PB Node Local Storage
- 716 PB Center-wide storage
- 4000 ft² foot print

Built by HPE

Olympus rack

- 128 AMD nodes
- 8,000 lbs
- Supports 400 KW



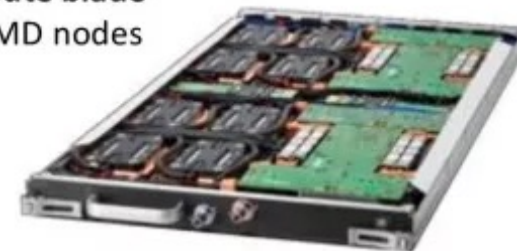
Powered by AMD

AMD node

- 1 AMD "Trento" CPU
- 4 AMD MI250X GPUs
- 512 GiB DDR4 memory on CPU
- 512 GiB HBM2e total per node
(128 GiB HBM per GPU)
- Coherent memory across the node
- 4 TB NVM
- GPUs & CPU fully connected with AMD Infinity Fabric
- 4 Cassini NICs, 100 GB/s network BW

Compute blade

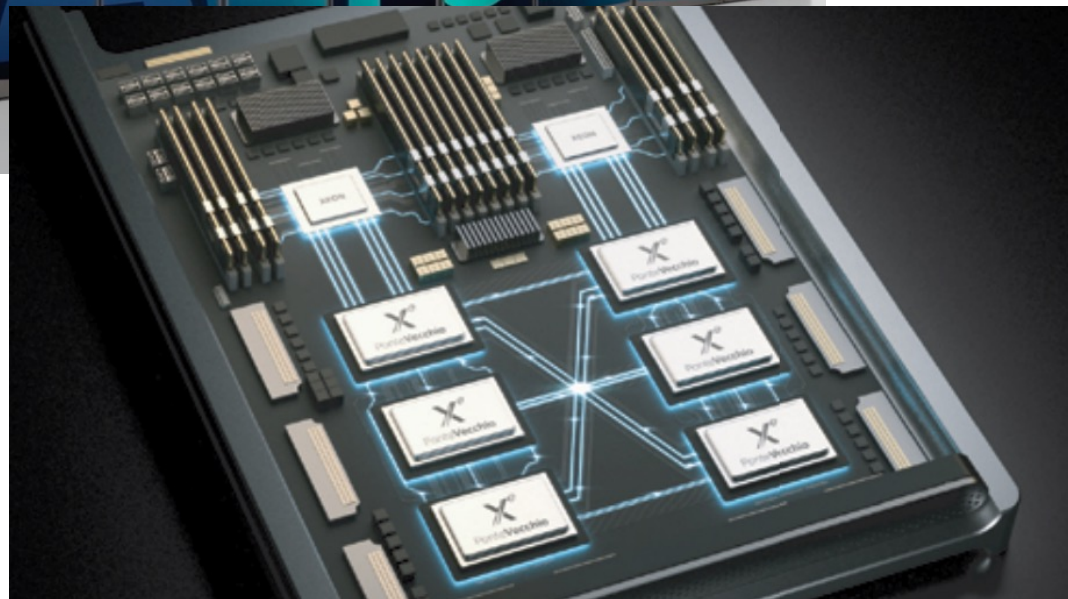
- 2 AMD nodes



All water cooled, even DIMMS and NICs



> 2 EFlops
38 MW !!!!
500 M USD



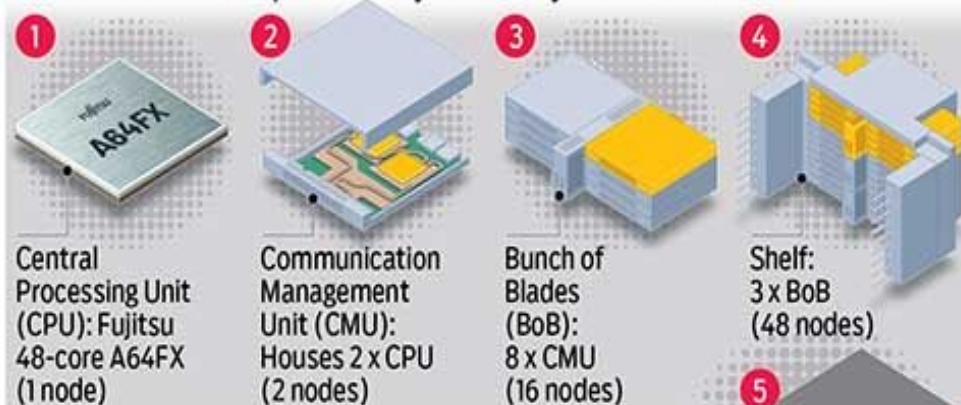
2

富岳

Fugaku

Supercomputer to seek Covid-19 cure

The world's fastest supercomputer, Japan's \$1.2 billion Fugaku, is to use its enormous power to try to identify treatments for Covid-19

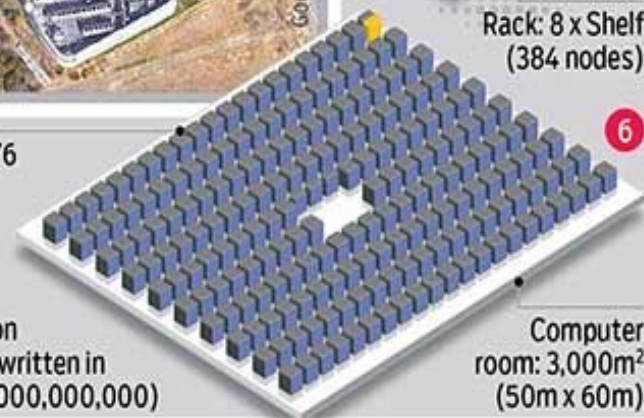


Jun 22, 2020: Top500 declares Fugaku world's fastest supercomputer – 2.8 times faster than previous record holder, IBM's Summit



System: 432 x Rack (158,976 nodes – comprising 396 racks of 384 nodes, plus 36 racks of 192 nodes)

Performance: 415.53 petaflops (415.53 quadrillion computations per second, written in numerals as: 415,530,000,000,000,000)



4
TOP 500
The List.

Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan

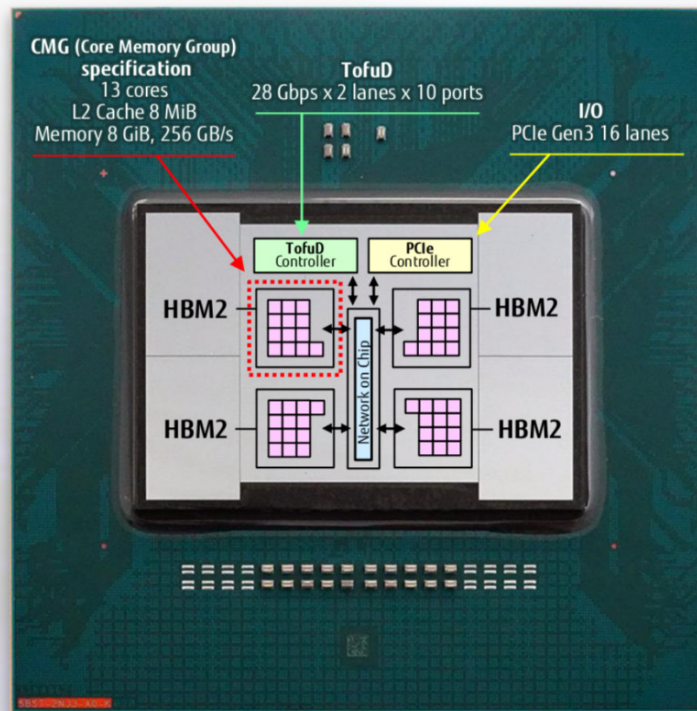
Nov'23



Fujitsu A64FX in Fugaku



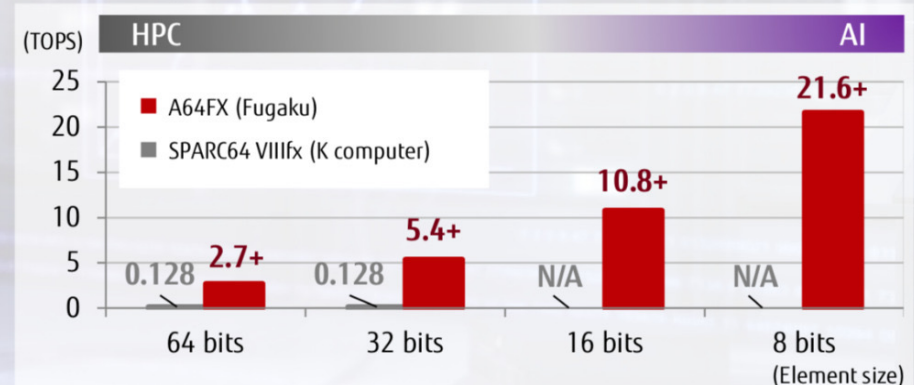
1. High-Performance Arm CPU A64FX in HPC and AI Areas



Architecture features

ISA	Armv8.2-A (AArch64 only) SVE (Scalable Vector Extension)	arm
SIMD width	512-bit	
Precision	FP64/32/16, INT64/32/16/8	
Cores	48 computing cores + 4 assistant cores (4 CMGs)	
Memory	HBM2: Peak B/W 1,024 GB/s	
Interconnect	TofuD: 28 Gbps x 2 lanes x 10 ports	

Peak performance (Chip level)



3

EuroHPC Supercomputers at the TOP10

TOP 500 Nov'23
The List.



LUMI #5

**3 EuroHPC
SUPERCOMPUTERS
RANKED AMONG THE
WORLD'S TOP 10
SUPERCOMPUTERS**

TOP 500



LEONARDO #6



MARENOSTRUM 5 #8



APereira, AJProença, SAC, 2025





LUMI at CSC, Finland



1 SYSTEM
380 Petaflop/s
PEAK PERFORMANCE

COMPUTING POWER EQUALS
1.5 MILLION
MODERN LAPTOP'S
CAPACITY

100%
HYDROELECTRIC
ENERGY

LUMI-C:
x86 Partition
Supplementary CPU partition:
over **262,000**
AMD EPYC CPU cores.

LUMI-K:
Container Cloud Service

LUMI-O:
Object Storage Service
30 PB
encrypted object storage
(Ceph) for storing, sharing
and staging data.

LUMI-Q:
Quantum Computing

High-speed interconnect

Possibility for combining
different resources within
a single run. HPE
Slingshot technology.

LUMI-G:
GPU Partition
Sustained performance
380
Pflop/s powered by AMD
Radeon Instinct™ MI250X GPUs.

LUMI-D:
Data Analytics Partition
Interactive partition with
32 TB
of memory and graphics GPUs for
data analytics and visualization.

LUMI-F:
Accelerated Storage
10 PB
Flash-based storage layer with
extreme I/O bandwidth of
2 TB/s and IOPS capability.

LUMI-P:
Lustre Storage
80 PB
parallel file system.



**Barcelona
Supercomputing
Center**

Centro Nacional de Supercomputación

MareNostrum 5 at BSC, Spain



6500 nodes

- 2x Intel Sapphire Rapids 8480+ 56c each (112 cores per node)
- 256 GB DDR5 (with 216 nodes with 1024GB)
- 960GB on NVMe storage (/scratch)
- 1x NDR200 shared by 2 nodes (SharedIO) (BW per node 100Gb/s)

1120 nodes

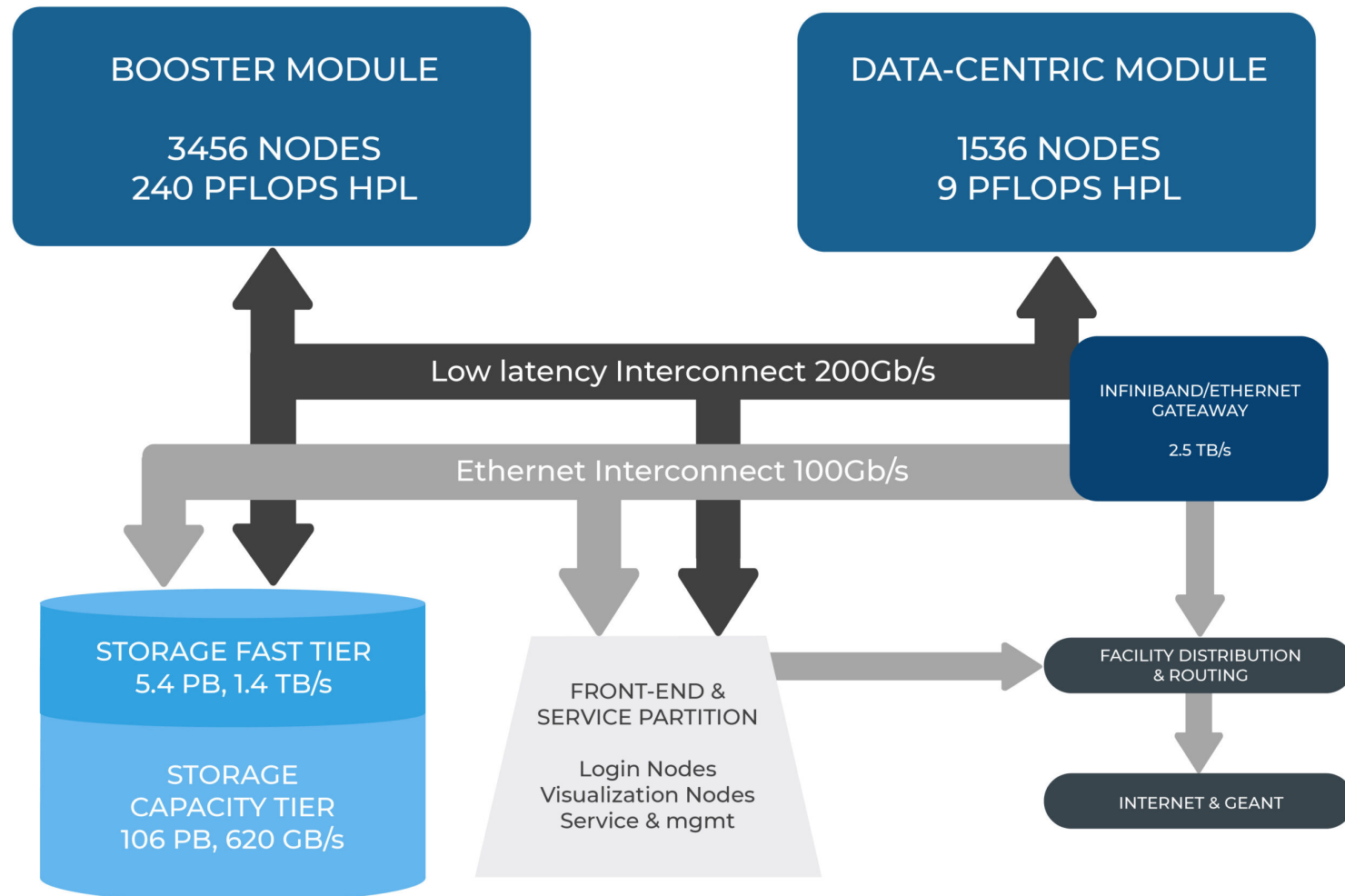
- 2x Intel Sapphire Rapids 8460Y+ 40c each (80 cores node)
- 512 GB of Main memory, using DDR5
- 4x Nvidia Hopper GPU with 64 HBM2 memory
- 480GB on NVMe storage (/scratch)
- 4x NDR200 (BW per node 800Gb/s)

APereira, AJProença, SAC, 2025





Leonardo at CINECA, Italy





Leonardo at CINECA, Italy



BOOSTER MODULE

3456 nodes

Features a **custom BullSequana X2135 “Da Vinci” blade**, composed of:

- 1 x CPU Intel Xeon 8358 32 cores, **Ice Lake** 2,6 GHz
- 512 (8 x 64) GB RAM DDR4 3200 MHz
- 4 x NVidia custom Ampere GPU 64GB HBM2
- 2 x NVidia HDR 2x100 Gb/s cards

Performance per node: 89,4 TFLOPs peak



6

Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy



DATA CENTRIC MODULE

1536 nodes

Features a **BullSequana X2610 compute blade**.

Each computing node is composed of:

- 2x Intel Sapphire Rapids, 56 cores, TDP 350 W
- 512 (16 x 32) GB RAM DDR5 4800 MHz
- 1 x NVidia HDR100 100 Gb/s card
- 8 TB NVM





PAGE CONTENTS

LUMI

LEONARDO

MARENOSTRUM 5

MELUXINA

KAROLINA

DISCOVERER

VEGA

DEUCALION

JUPITER

}
in TOP10

DEUCALION

Deucalion is a petascale EuroHPC supercomputer located in Guimarães, Portugal. It is supplied by Fujitsu Technology Solutions combining a Fujitsu PRIMEHPC (ARM partition) and Atos Bull Sequana (x86 partitions). Deucalion is hosted by [MACC](#).



FCT

7.22 petaflops

Sustained performance

10.00 petaflops

Peak performance

Compute partitions:

ARM Partition: 1632 nodes, 3.8 PFLOps ; x86 Partition: 500 nodes, 1,62 PFLOps ; Accelerated: 33 nodes, 1,72 PFLOps

Central Processing Unit (CPU):

A64FX (ARM partition), AMD EPYC (x86 partitions)

Graphics Processing Unit (GPU):

NVidia Ampere

Storage capacity:

430 TB High-speed NVMe partition, 10.6 PB high-speed based Parallel File System partition.

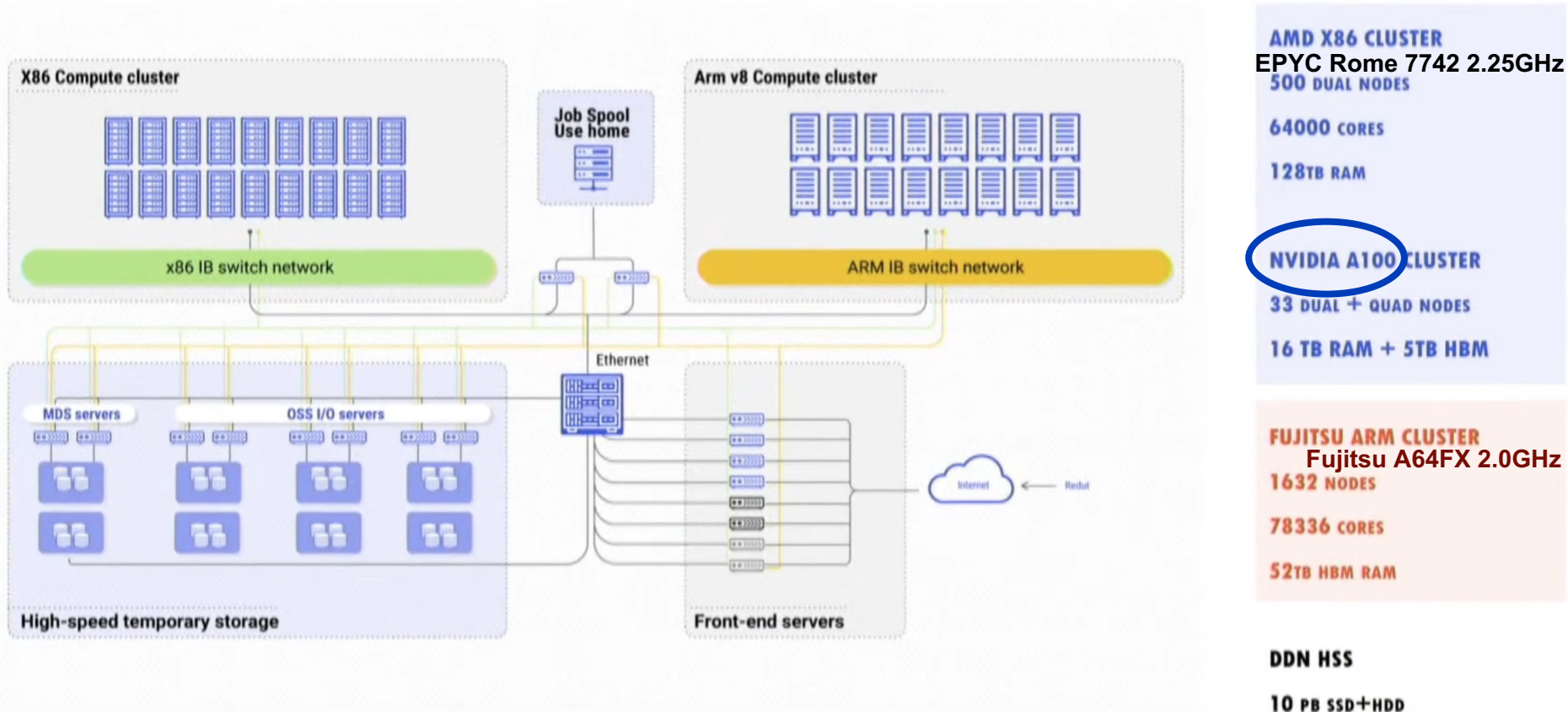
Applications:

Traditional Computational, AI, Big Data

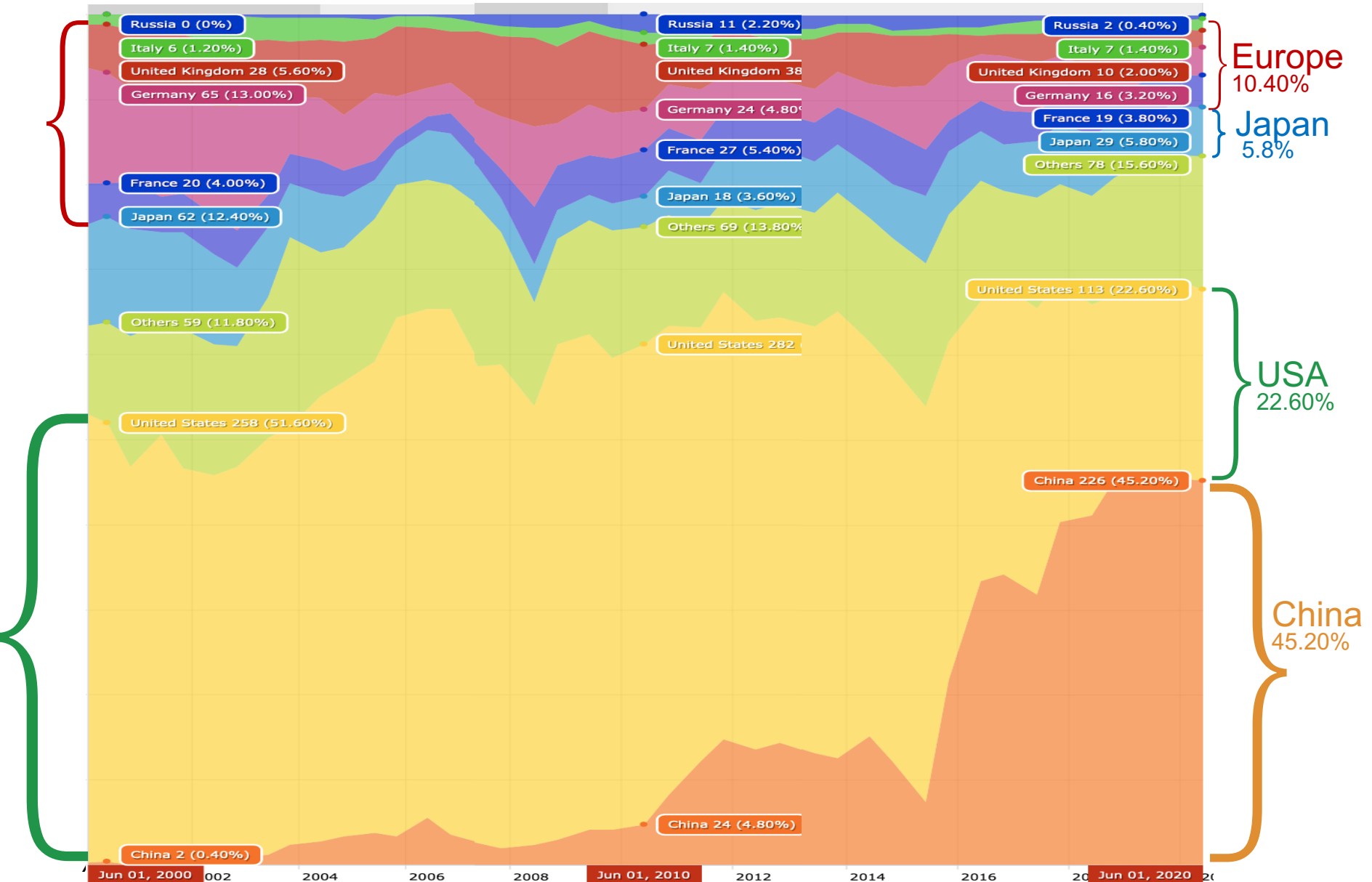
The Deucalion in MACC



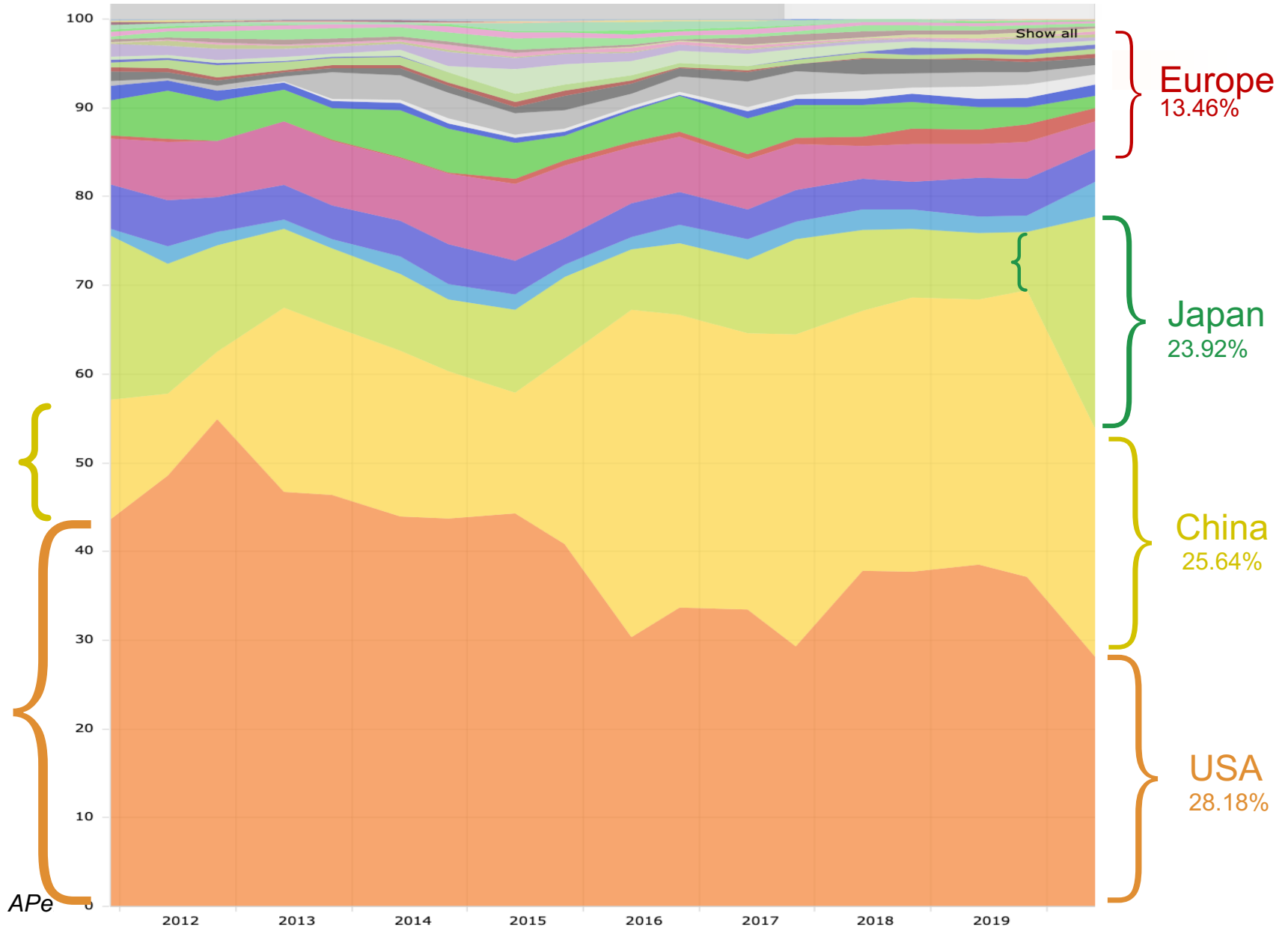
DEUCALION OVERALL ARCHITECTURE



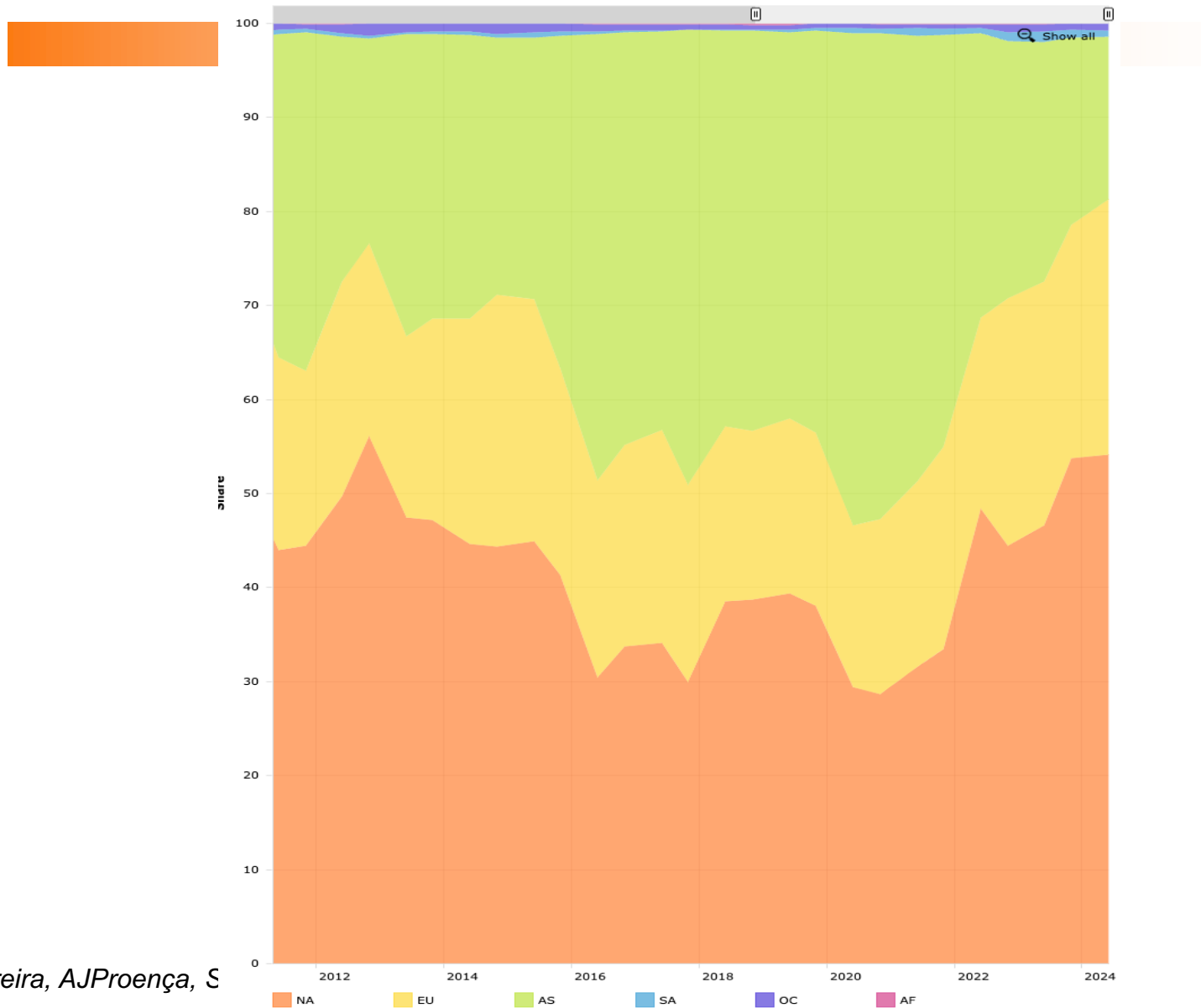
Country distribution over 20 years: from 2000 to 2020, #systems



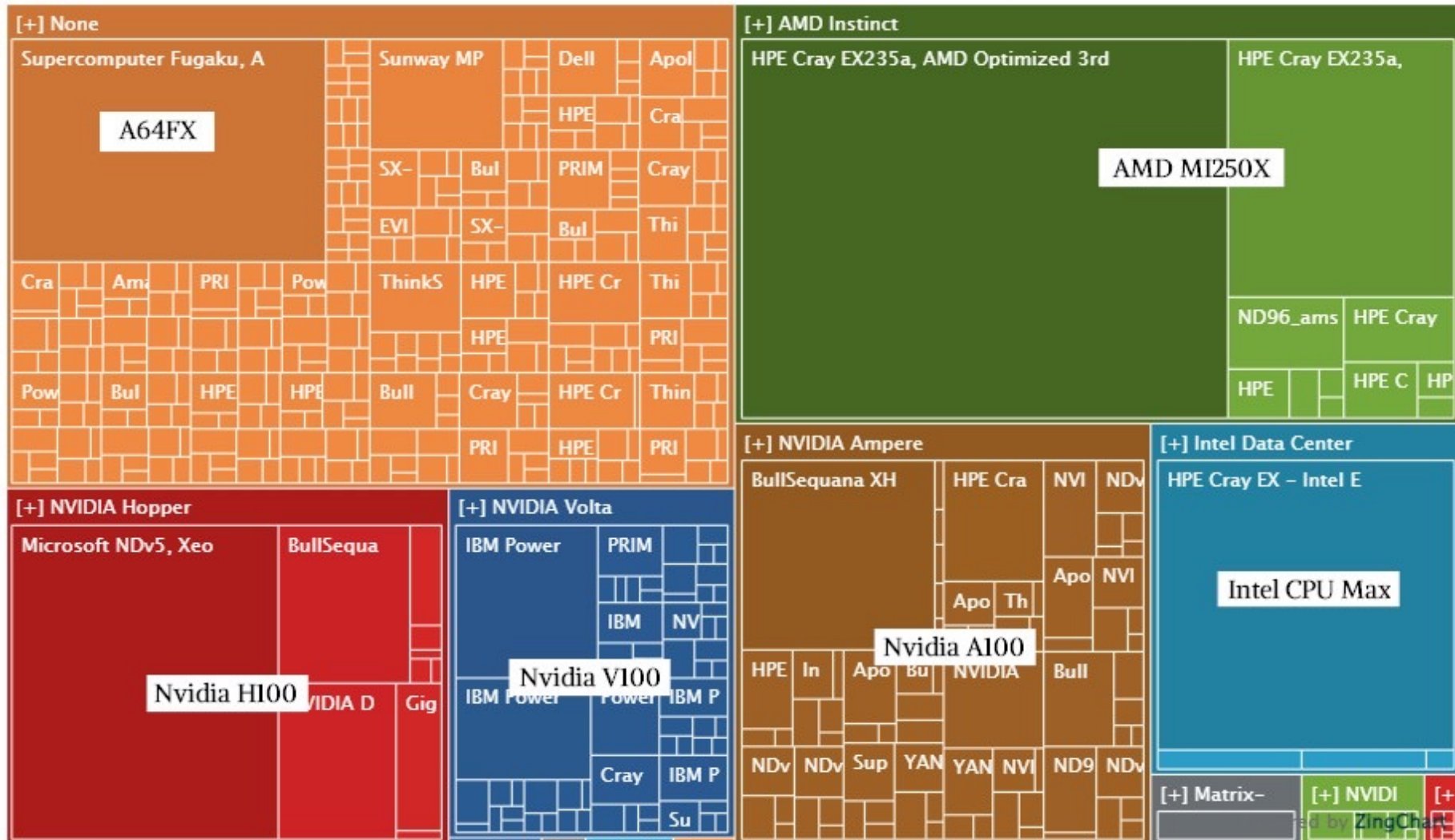
Country distribution over 20 years: from 2000 to 2020, *aggregate performance*



Continent distribution in 2024 aggregate performance



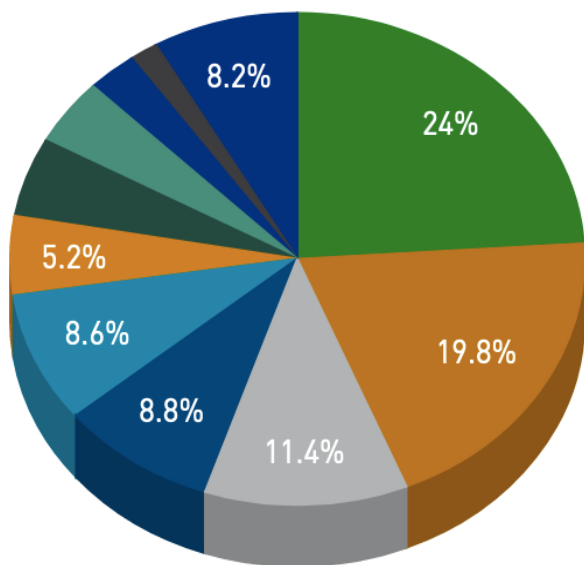
Distribution by processor/accelerator family: #systems & performance (Nov'23)



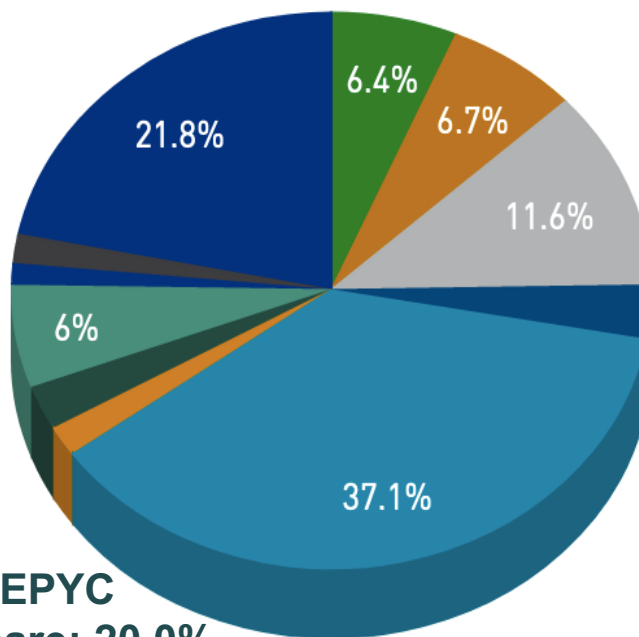


Processor Generation System Share

Processor Generation Performance Share



Intel Xeon
System share: 70.5%
Performance share: 27.8%



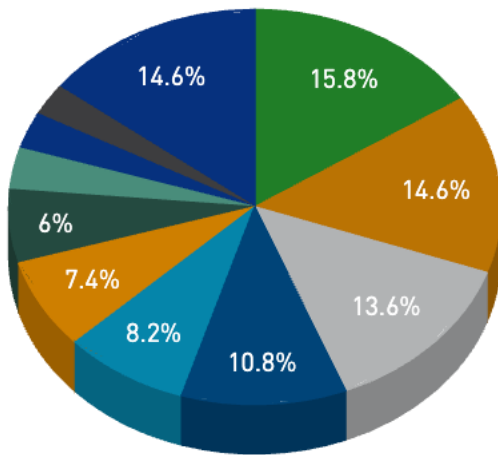
AMD EPYC
System share: 20.0%
Performance share: 48.7%

- Xeon Gold
- Xeon Gold 62xx (Cascade Lake)
- AMD Rome
- Xeon Platinum 82xx (Sagefield)
- AMD Milan
- Intel Xeon E5 (Broadwell)
- Xeon Platinum
- Xeon® Platinum 83xx (Sagefield)
- Intel Xeon E5 (Haswell)
- Intel Xeon Phi
- Others

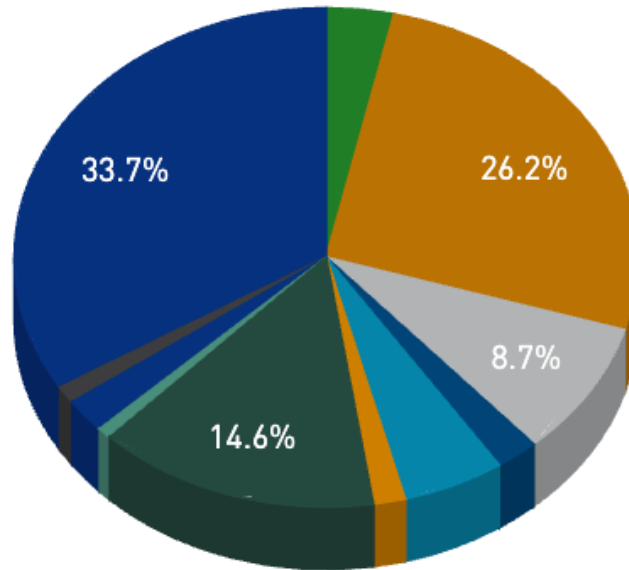
Processor generation share Jun'24



Processor Generation System Share



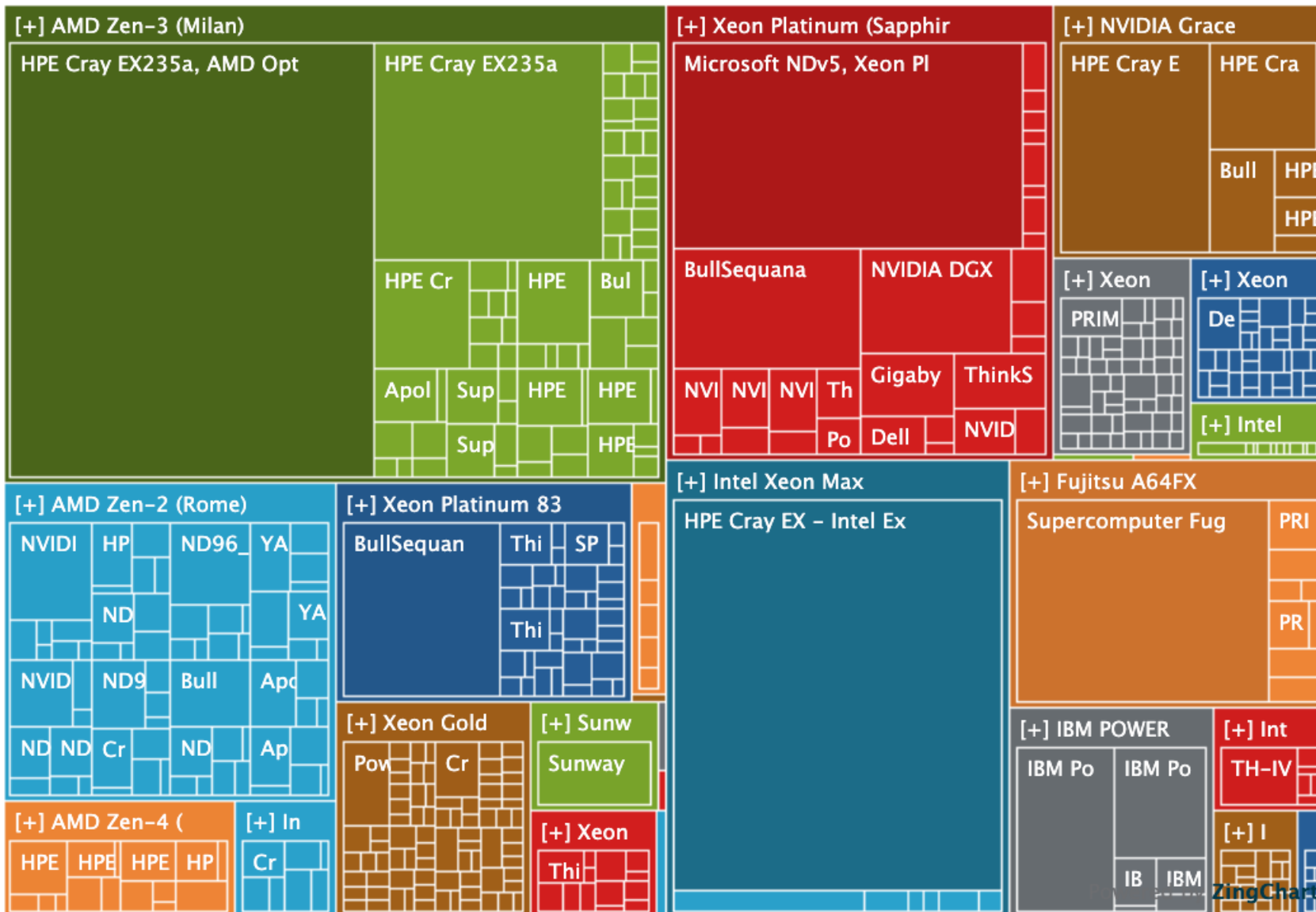
Processor Generation Performance Share



- Xeon Gold 62xx (Cascade Lake)
- AMD Zen-3 (Milan)
- AMD Zen-2 (Rome)
- Xeon Gold (Skylake)
- Xeon Platinum 83xx (Ice Lake)
- Xeon Platinum 82xx (Cascade Lake)
- Xeon Platinum (Sapphire Rapids)
- Intel Xeon E5 (Broadwell)
- AMD Zen-4 (Genoa)
- Xeon Platinum (Skylake)
- Others

Processor distribution

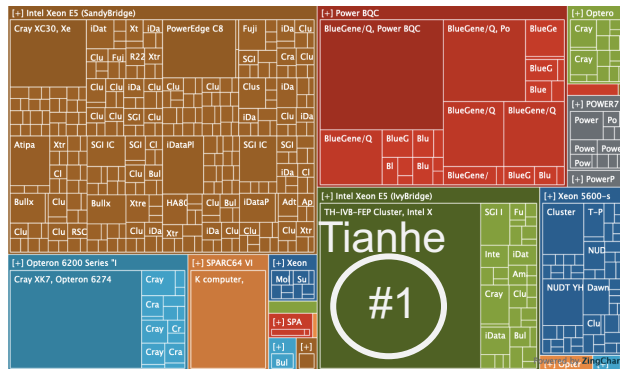
Jun'24



Architecture family of key #1's: Nov'13 to Nov'23

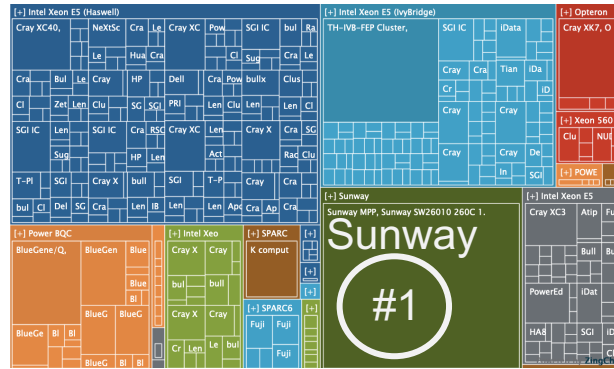


Nov'13



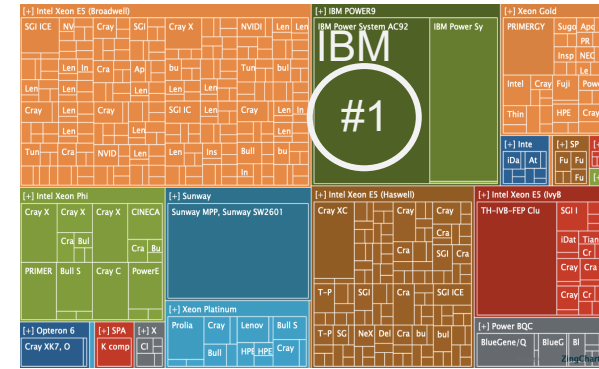
with Xeon Phi

Jun'16



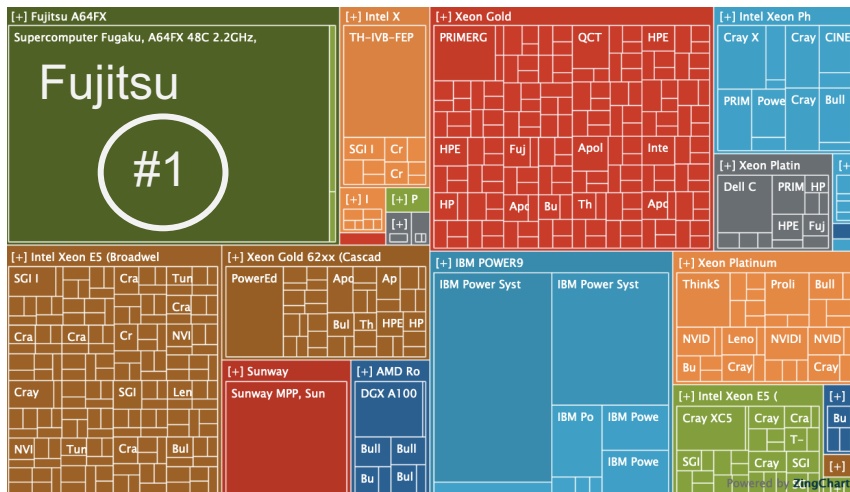
without any accelerator

Jun'18



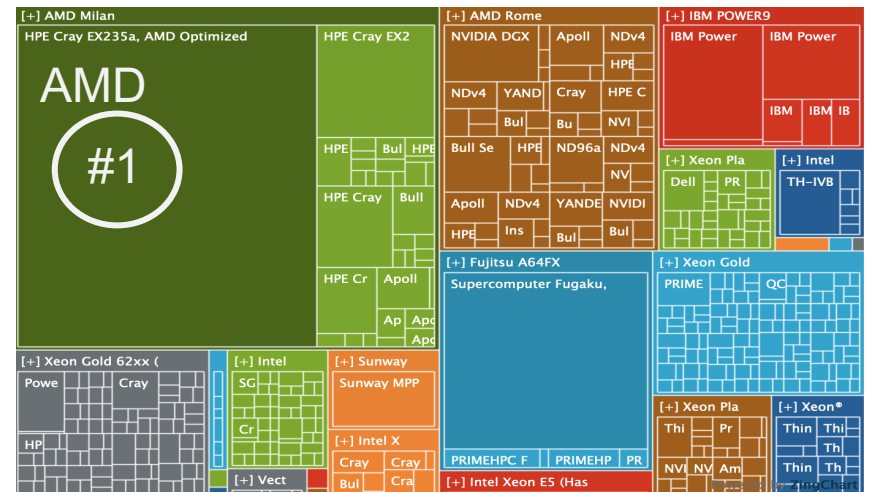
with Nvidia Volta

Jun'20



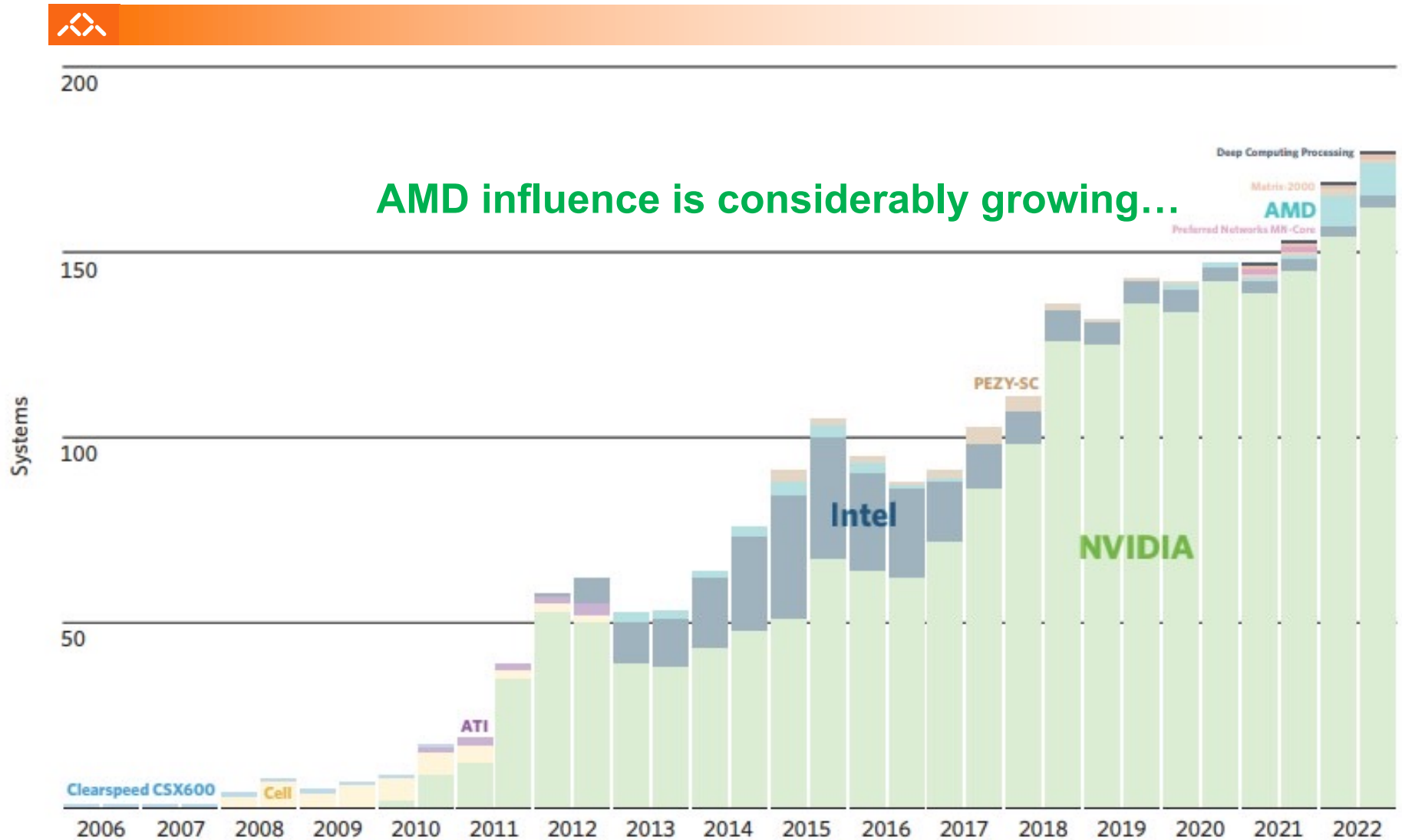
without any accelerator

Jun'22



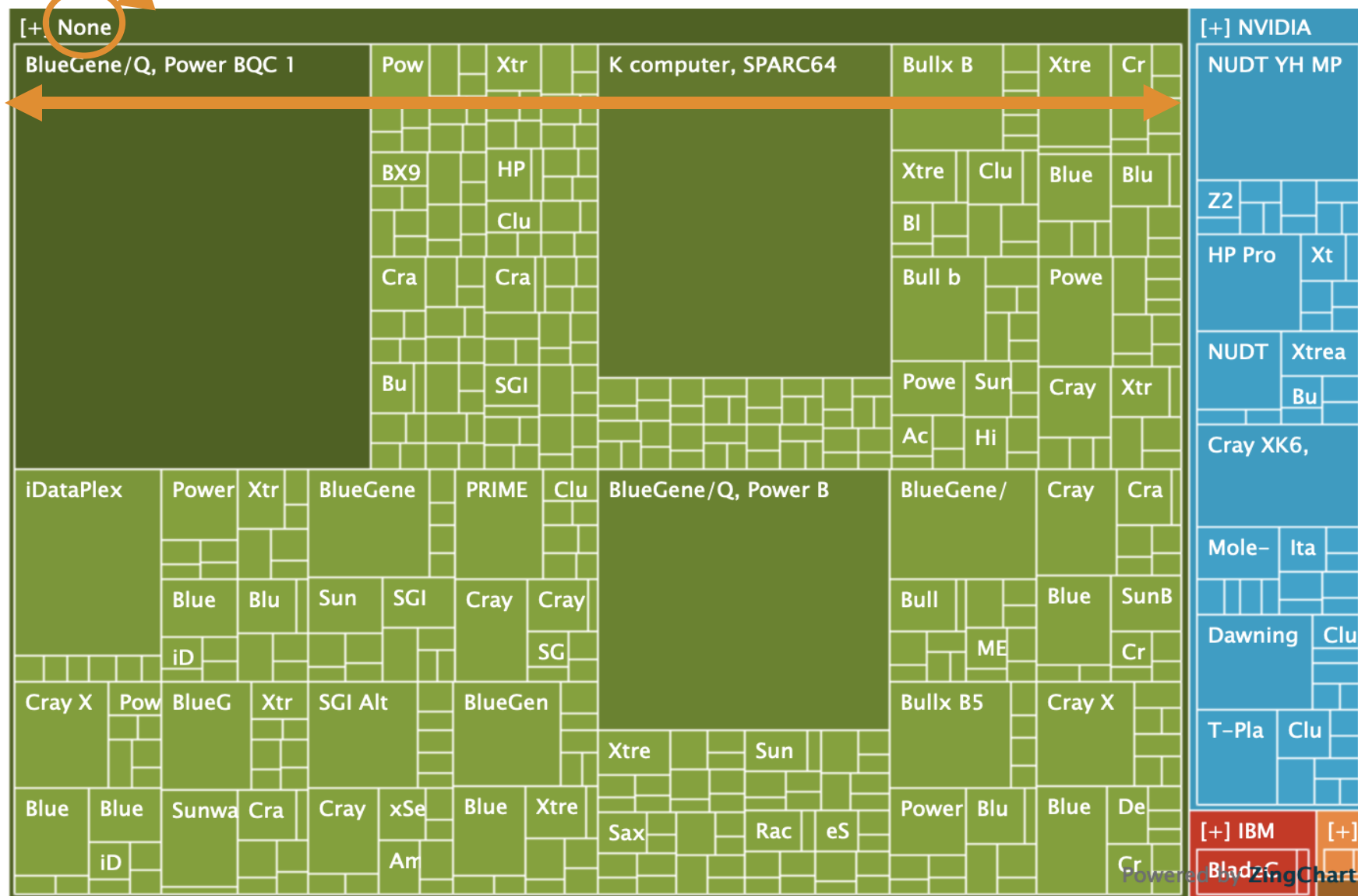
with AMD MI250x

Accelerator families 2006 to 2022



Accelerator family distribution

Jun'12



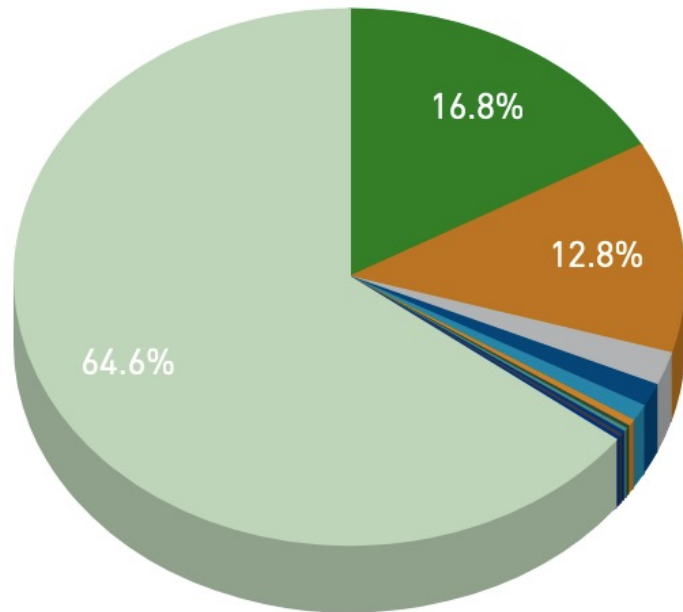


Accelerator family distribution

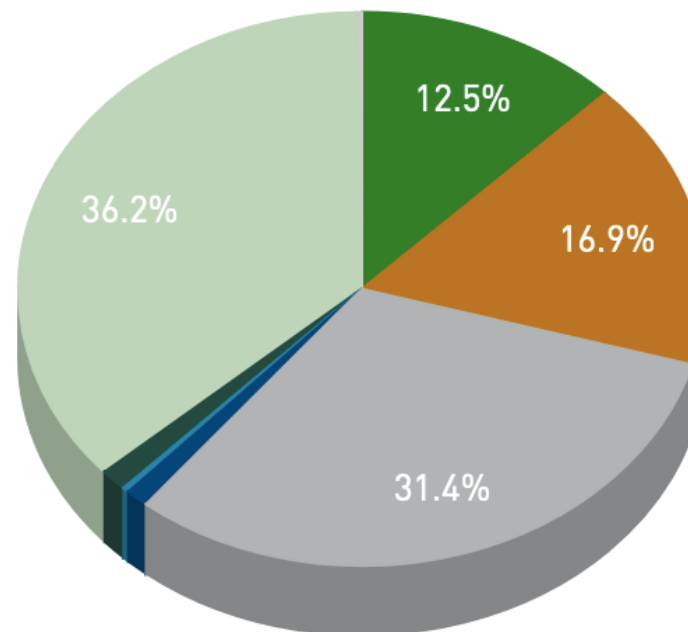
Nov'23



Accelerator/CP Family System Share



Accelerator/CP Family Performance Share



- NVIDIA Volta
- NVIDIA Ampere
- AMD MI250x
- NVIDIA Pascal
- NVIDIA Kepler
- Matrix-2000
- Others
- Other



1. TOP500

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The list ranks computers in terms of energy efficiency, typically measured as LINPACK FLOPS per watt.

About the Green500 List

The Green500 list ranks the top 500 supercomputers in the world by energy efficiency. The focus of performance-at-any-cost computer operations has led to the emergence of supercomputers that consume vast amounts of electrical power and produce so much heat that large cooling facilities must be constructed to ensure proper performance. To address this trend, the Green500 list puts a premium on energy-efficient performance for sustainable supercomputing.

The inaugural Green500 list was announced on November 15, 2007 at SC|07. As a complement to the TOP500, the unveiling of the Green500 ushered in a new era where supercomputers can be compared by performance-per-watt.

While the selection of any power-performance metric will be controversial, we currently opt for "FLOPS-per-Watt" given that it has already become a widely used metric in the community and for



Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	Power (kW)	Energy Efficiency (GFlops/watts)
1	189	JEDI - BullSequana XH3000, Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, ParTec/EVIDEN EuroHPC/FZJ Germany	19,584	4.50	67	72.733
2	128	Isambard-AI phase 1 - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE University of Bristol United Kingdom	34,272	7.42	117	68.835
3	55	Helios GPU - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cyfronet Poland	89,760	19.14	317	66.948
4	328	Henri - ThinkSystem SR670 V2, Intel Xeon Platinum 8362 32C 2.8GHz, NVIDIA H100 80GB PCIe, Infiniband HDR, Lenovo Flatiron Institute United States	8,288	2.88	44	65.396
5	71	preAlps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	81,600	15.47	240	64.381
6	299	HoreKa-Teal - ThinkSystem SD665-N V3, AMD EPYC 9354 32C 3.25GHz, Nvidia H100 94Gb SXM5, Infiniband NDR200, Lenovo Karlsruher Institut für Technologie (KIT) Germany	13,616	3.12	50	62.964
7	54	Frontier TDS - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	120,832	19.20	309	62.684

10 systems
Nov'22

Still Nvidia dominated...



1. TOP500

- a) TOP10 lists from Nov'17 to Nov'23
- b) Analysis of some relevant systems and architectures
- c) Country distribution over the past 25 years
- d) PU chip technology evolution in the past 25 years and since last year
- e) Evolution of the accelerators since they were available

2. GREEN500

- a) TOP5 list from Nov'19 to Nov'22
- b) Analysis of some relevant systems and architectures

3. HPCG

- a) HPCG vs. HPL: an overview
- b) TOP10 in Nov'22

4. HPL-AI

- a) High-performance Linpack (HPL) and artificial intelligence (AI) workloads

HPCG benchmark



HPCG is a self-contained C++ program with MPI and OpenMP support that measures the performance of basic operations in a unified code:

HPCG benchmark

From Wikipedia, the free encyclopedia

- Sparse matrix-vector multiplication
- Vector updates
- Global dot products
- Local symmetric Gauss-Seidel smoother
- Sparse triangular solve (part of Gauss-Seidel smoother)

The **HPCG (high performance conjugate gradient) benchmark** is a [supercomputing benchmark](#) test proposed by Michael Heroux from [Sandia National Laboratories](#), and [Jack Dongarra](#) and Piotr Luszczek from the [University of Tennessee](#).^{[1][2]} It is intended to model the [data access](#) patterns of real-world [applications](#) such as [sparse matrix](#) calculations, thus testing the effect of limitations of the [memory subsystem](#) and internal [interconnect](#) of the supercomputer on its computing performance.^[3] Because it is internally [I/O bound](#), HPCG testing generally achieves only a tiny fraction of the peak [FLOPS](#) of the computer.^[4]

HPCG is intended to complement benchmarks such as the [LINPACK benchmarks](#) that put relatively little stress on the internal interconnect.^[5] The source of the HPCG benchmark is available on [GitHub](#).^[6]



TOP500: HPCG vs. HPL



HPCG Benchmark

The High Performance Conjugate Gradients (HPCG) Benchmark project is an effort to create a new metric for ranking HPC systems. HPCG is intended as a complement to the High Performance LINPACK (HPL) benchmark, currently used to rank the TOP500 computing systems. The computational and data access patterns of HPL are still representative of some important scalable applications, but not all. HPCG is designed to exercise computational and data access patterns that more closely match a different and broad set of important applications, and to give incentive to computer system designers to invest in capabilities that will have impact on the collective performance of these applications.

HPCG is a complete, stand-alone code that measures the performance of basic operations in a unified code:

- Sparse matrix-vector multiplication.
- Vector updates.
- Global dot products.
- Local symmetric Gauss-Seidel smoother.
- Sparse triangular solve (as part of the Gauss-Seidel smoother).
- Driven by multigrid preconditioned conjugate gradient algorithm that exercises the key kernels on a nested set of coarse grids.
- Reference implementation is written in C++ with MPI and OpenMP support.



Top systems

Jun'24



Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	HPCG (TFlop/s)
1	4	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	16004.50
2	1	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,699,904	1,206.00	14054.00
3	2	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	5612.60
4	5	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	4586.95
5	6	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	1,305,600	270.00	3671.32

NVIDIA
advantage?

European Supercomputing



EuroHPC
Joint Undertaking

APereira, AJProença, SAC, 2025



European Supercomputing



EuroHPC
Joint Undertaking



NOV 2023	TOP500	Green500
LUMI	#5	#7
LEONARDO	#6	#18
MARENOSTRUM 5	#8	#6
MELUXINA	#71	#27
KAROLINA	#113	#25
DISCOVERER	#166	#216
VEGA	#198	#253



EuroHPC
Joint Undertaking

EuroHPC supercomputers: *their locations*

